INTEGRATING SUSTAINABILITY INTO BUSINESS FUNCTIONS

Sustainable Marketing & Branding

Developing and communicating sustainable products and services — without overclaiming

Since 27 September 2026, calling a product “eco-friendly” without proof is illegal across the EU. India’s regulator issued equivalent guidelines in October 2024.

Marketing used to be the one function allowed to describe sustainability in adjectives. Today it has to describe it in evidence — and that changes how products are designed, priced, and sold.

BY THE END OF THIS LECTURE

You will be able to...



Design

Apply the green marketing mix — product, price, place, promotion — to a sustainable offering.



Calculate

Frame a sustainable product's price premium as lifetime cost and avoided emissions.



Evaluate

Test any environmental claim against five principles and current Indian and EU rules.



Connect

Explain what marketing needs from every other business function to keep its promises.

TODAY'S JOURNEY

Marketing: The Function That Speaks for All the Others

Sustainability is integrated function by function. Marketing sits at the customer interface — so it is both the loudest voice and the most exposed to legal and reputational risk.



01

Developing Sustainable Offerings

What makes an offering genuinely more sustainable — and how to price and position it.



02

Communicating Credibly

Claims, labels, and storytelling that customers and regulators can both trust.



03

Greenwashing & Regulation

The seven sins, and the rules now in force in India, the EU, the UK, and the US.



04

Purpose-Driven Branding

Turning proof into brand value — and what Patagonia and H&M teach us.

SECTION 1 OF 4

Developing Sustainable Offerings

Marketing starts before the campaign: with what is actually being sold.

The Green Marketing Mix: The Four Ps, Reimagined

The classic marketing mix still applies — but each P carries a sustainability decision.



Product

Design for lower impact and longer life; substantiate improvements with LCA data rather than adjectives.



Price

Price on lifetime cost, not sticker price; decide deliberately whether to charge a premium, match, or move to a service model.



Place

Low-carbon distribution and availability where customers already shop, plus refill or take-back points that make the sustainable choice convenient.



Promotion

Honest, specific communication that helps customers choose and use the product well — the subject of Sections 2 and 3.

SECTION 1 · PRODUCT DESIGN

Design Decisions Lock In the Impact

The LCA lecture showed where impacts sit; design is where they are decided — long before any advertisement is written.

Design for Durability & Repair	A longer life spreads the embedded impact of making the product over more years of use — often the single biggest lever.
Design for Disassembly & Recycling	Mono-materials, removable batteries, and labelled components keep materials in the technical cycle instead of the landfill.
Design for Efficient Use	Where the use phase dominates (appliances, vehicles), energy and water in use matter far more than packaging — Scope 3 Category 11.
Design for Less Material	Lightweighting, concentrates, and refills cut material and transport impact per use — and are easy to explain honestly.

The Attitude–Behaviour Gap

Why “customers say they care” does not automatically become “customers buy it”.

The Gap

Surveys routinely find far more people intend to buy sustainable products than actually do — researchers call this the attitude–behaviour (or intention–action) gap (Carrington, Neville & Whitwell, 2010).

Why It Happens

Price premiums, limited availability, doubt about whether claims are true, habit, convenience, and unclear personal benefit all stand between intention and purchase.

What It Means for Marketers

Do not rely on values alone. Make the sustainable option the affordable, convenient, default, and verifiable one — and show the personal payoff.

SECTION 1 · WORKED EXAMPLE

Selling on Lifetime Cost: A 5-Star vs. 3-Star Appliance

GIVEN

- Product A (3-star): ₹40,000; uses 1,200 kWh per year
- Product B (5-star): ₹48,000; uses 800 kWh per year
- Tariff ₹8/kWh; 10-year life; grid factor 0.727 $kgCO_2e$/kWh (illustrative)

FORMULA

Lifetime Cost = Purchase Price + (Annual kWh × Tariff × Years)

STEP-BY-STEP CALCULATION

1. Product A: ₹40,000 + (1,200 × ₹8 × 10) = ₹40,000 + ₹96,000 = ₹1,36,000
2. Product B: ₹48,000 + (800 × ₹8 × 10) = ₹48,000 + ₹64,000 = ₹1,12,000
3. Saving with B: ₹1,36,000 − ₹1,12,000 = ₹24,000; the ₹8,000 premium is repaid in 8,000 ÷ 3,200 = 2.5 years
4. Emissions avoided: 400 kWh × 0.727 $kgCO_2e$ × 10 years ≈ 2,908 kg ≈ 2.9 tCO_2e

₹24,000 + 2.9 t CO_2e

Reframing the sticker-price premium as a lifetime saving is both the most honest and the most persuasive message. (Figures are illustrative, not product data.)

Beyond the Product: Selling the Service

Some of the strongest sustainability offers change what is sold, not just how it is made.



Repair & Maintenance

Lifetime repair services extend product life and deepen loyalty — Patagonia's repair and Worn Wear programmes are the best-known example.



Refill & Reuse

Refillable packaging and deposit-return systems remove single-use material entirely, if convenience keeps pace.



Resale & Take-Back

Buying back used products for refurbishment or resale keeps value and materials circulating — the circular economy at the till.



Leasing & Subscription

Selling use instead of ownership aligns the seller's profit with durability, because the seller keeps the asset.

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

Why is it usually more persuasive to market a 5-star appliance on lifetime cost than on its sticker price?

A. Sticker prices are not allowed in sustainability advertising

B. The upfront premium is repaid through savings, so lifetime cost reframes the premium as a saving ✓

C. Lifetime cost hides the real price from the customer

D. Star labels only matter for exported products

ANSWER & WHY

B. The upfront premium is repaid through savings, so lifetime cost reframes the premium as a saving

In the worked example the ₹8,000 premium is repaid in 2.5 years and saves ₹24,000 over ten years. Showing that is honest, verifiable, and speaks to the customer's own interest — the attitude–behaviour gap's biggest barrier is price.

SECTION 2 OF 4

Communicating Credibly

A claim is only as strong as the evidence a sceptical stranger could check.

SECTION 2 · THE STANDARD

Five Principles of a Credible Claim

Distilled from regulator guidance in India, the EU, and the UK — a checklist to run before anything is published.



Specific

Say exactly what and how much — “60% recycled content” — not “eco” or “green”.



Substantiated

Keep an evidence file behind every claim; use independent verification wherever possible.



Relevant

Talk about impacts that matter, not legal baselines or trivial attributes.



Complete

Consider the full life cycle and disclose important trade-offs rather than hiding them.



Proportionate

Claim no more than the evidence supports — including comparisons and future targets.

Vague vs. Verifiable: Rewriting the Same Message

At-Risk Claim	Credible Version
“Our bottle is eco-friendly.”	Made from 60% recycled PET, verified by an independent auditor; the other 40% is virgin resin.
“Carbon-neutral shipping.”	We cut delivery emissions per parcel by 18% since 2023 (third-party verified); removal projects are funded separately and reported separately.
“Good for the planet.”	Uses 30% less electricity than the 3-star equivalent, per the BEE energy label.
“Net zero soon.”	Targets validated by the Science Based Targets initiative, with interim milestones and annual progress reporting.

Ecolabels: Three Types, Three Levels of Trust

The ISO 14020 family classifies environmental labels by who checks the claim.



Type I · ISO 14024

Third-party programmes with published criteria and independent verification — for example the EU Ecolabel or Germany's Blue Angel.



Type II · ISO 14021

Self-declared claims by the manufacturer. Legitimate, but only if specific, verifiable, and not misleading.



Type III · ISO 14025

Environmental Product Declarations: LCA-based, quantified, third-party verified — data, not a pass/fail badge.



India: BEE & Ecomark

BEE Star Labels rate appliance energy efficiency; Ecomark is India's ecolabelling scheme for products.

◆ INTERACTIVE

CLASSIFY THE CLAIM

Credible or At Risk?

For each claim, decide whether it is credible as written or at risk under current guidance.

1. “Eco-friendly” printed on packaging, with no explanation
2. “Contains 60% post-consumer recycled plastic,” audited by a third party
3. “Carbon-neutral product,” achieved purely through carbon offsets
4. “25% lower cradle-to-gate CO_2e than our 2023 model,” per a reviewed LCA
5. Green leaf imagery and “good for the planet” on a plastic-heavy product
6. “Uses 30% less electricity than the 3-star equivalent,” per BEE label data

CREDIBLE	AT RISK
2	1
4	3
6	5

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

Which type of ecolabel is an Environmental Product Declaration (EPD)?

A Type I (ISO 14024) — third-party ecolabel programme

B Type II (ISO 14021) — self-declared claim

C Type III (ISO 14025) — quantified, LCA-based, third-party verified declaration ✓

D None of the ISO ecolabel types

ANSWER & WHY

C. Type III (ISO 14025) — quantified, LCA-based, third-party verified declaration

EPDs are Type III declarations: they report quantified life cycle data (from an LCA, as in our previous lecture) and are third-party verified, but they do not rank a product as good or bad.

SECTION 3 OF 4

Greenwashing & Regulation

The line between persuasion and deception is now written into law.

The Seven Sins of Greenwashing (TerraChoice, 2007)

Hidden Trade-off

Green on one attribute while ignoring larger impacts elsewhere.

No Proof

A claim with no accessible evidence or credible certification.

Vagueness

Poorly defined terms such as “all-natural” or “eco”.

Irrelevance

True but unimportant — or already required by law.

Lesser of Two Evils

Green within a category that is harmful overall.

Fibbing

Claims that are simply false.

Worshipping False Labels

Fake seals or implied endorsements that do not exist.

Four Jurisdictions, One Direction of Travel

Different tools, same message: claims must be specific, evidenced, and not misleading.



India

CCPA Greenwashing Guidelines (15 Oct 2024): generic terms need qualifiers and substantiation; aspirational claims need an actionable plan.



European Union

Directive (EU) 2024/825, applying from 27 Sep 2026: bans generic claims without proof, offset-based neutrality claims, and self-made labels.



United Kingdom

CMA Green Claims Code (2021); in 2022 the CMA opened a fashion-sector investigation covering ASOS, Boohoo, and others.



United States

FTC Green Guides (last revised 2012): guidance rather than statute, enforced through deceptive-advertising law, state actions, and class suits.

SECTION 3 · INDIA

India’s CCPA Greenwashing Guidelines, 2024

Issued by the Central Consumer Protection Authority on 15 October 2024, complementing the 2022 misleading-advertisements guidelines.

Who Is Covered	Manufacturers, service providers, traders, advertising agencies, and endorsers — every party that makes or amplifies an environmental claim.
Generic Terms Need Backing	Words such as “green”, “eco-friendly”, “sustainable”, “natural”, “pure” or “carbon-neutral” cannot be used without adequate, accurate, accessible qualifiers and substantiation.
The Evidence Standard	Claims need verifiable evidence — independent third-party certification is accepted — and comparative claims must rest on relevant data disclosed to consumers.
Aspirational Claims	Future-facing claims such as “net zero by 2050” are acceptable only when clear, actionable plans exist to deliver them.

The EU's Green Claims Rules: Timeline

27 Mar 2024

Directive (EU) 2024/825 enters into force, amending EU consumer-protection law.

27 Mar 2026

Deadline for Member States to write the directive into national law.

27 Sep 2026

Rules apply to traders, with no transition period for products already on the market.

After Sep 2026

National authorities and courts enforce; the separate Green Claims Directive proposal has been paused since 2025.

What the EU Now Prohibits

The directive amends the Unfair Commercial Practices Directive — these practices are now unfair by definition.

Generic Environmental Claims	“Eco-friendly”, “green”, “climate-friendly” and similar are banned unless the trader can show recognised excellent environmental performance relevant to the claim.
Offset-Based Neutrality Claims	Saying a product is climate neutral, or has a reduced or positive impact, because of greenhouse-gas offsetting is prohibited at product level.
Self-Made Sustainability Labels	Labels must rest on a certification scheme or be established by public authorities; a company’s own badge no longer counts.
Unsupported Future Claims	Claims about future performance need a detailed, time-bound implementation plan, verified by an independent expert, with the findings made available.

◆ INTERACTIVE

CASE STUDY

When a “Conscious” Label Meets a Regulator: H&M



H&M’s “Conscious Choice” collection and Higg Index scorecards (2019–2022)

Regulators in the Netherlands and Norway challenged H&M’s use of labels such as “Conscious” and of Higg Materials Sustainability Index scores, arguing that the claims were unclear and insufficiently substantiated.

The Regulatory Response

Norway’s Consumer Authority criticised the Conscious collection in 2019 and warned in June 2022 that Higg-based marketing could be misleading; the Dutch ACM found unclear, insufficiently substantiated claims.

The Outcome

In September 2022 H&M removed “Conscious Choice” indicators from its online shops worldwide, dropped a “more sustainable materials” heading, and donated €500,000 to sustainability organisations.

The Industry Ripple

The Sustainable Apparel Coalition paused its consumer-facing Higg transparency programme on 27 June 2022, and the UK CMA opened its fashion-sector probe in July 2022.

The Balance

H&M said the concerns did not involve “false information”, and US class actions have had mixed outcomes: the lesson is about clarity and substantiation, not a finding of fraud.

SECTION 3 · THE OPPOSITE RISK

Greenhushing: When Fear Silences Real Progress

Regulation punishes overclaiming — but staying silent has its own cost.

What It Is	Companies deliberately under-communicate real sustainability progress to avoid scrutiny or backlash — “greenhushing”.
Why It Is a Problem	Customers, investors, and suppliers cannot reward progress they never hear about, and best practice does not spread.
The Remedy: Say Less, Prove More	Communicate specific, modest, evidenced progress — including what is not yet solved. Precision is protection against both greenwashing and greenhushing.

◆ INTERACTIVE

THINK & SHARE

“Your product’s emissions per unit fell 18%, but total company emissions rose because sales grew. What can you truthfully say in an advertisement?”

Think about intensity versus absolute claims — and what a regulator would ask you to prove.

⏱ 3 MIN

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

From 27 September 2026, which claim is banned as an unfair practice across the EU?

A “Contains 60% recycled material,” verified by an independent auditor

B A product-level “carbon neutral” claim based on greenhouse-gas offsetting ✓

C Publishing an LCA-based Environmental Product Declaration

D Any mention of sustainability in advertising

ANSWER & WHY

B. A product-level “carbon neutral” claim based on greenhouse-gas offsetting

Directive (EU) 2024/825 prohibits product-level climate-neutral claims that rest on offsetting. Specific, verified claims and EPDs remain allowed — which links directly back to our carbon-markets lecture on offset quality.

SECTION 4 OF 4

Purpose-Driven Branding

Trust is the asset: earned by deeds first, and only then by messages.

Three Tests for Purpose-Driven Branding

Purpose is not a slogan; it is a promise the whole company has to be able to keep.

Authentic

The commitment connects to the company's core business — not to a side campaign that leaves the main product line untouched.

Material

It addresses the impacts that actually dominate the footprint (the hotspots found by LCA and Scope 3 analysis), not just the easy ones.

Verifiable

Progress is measured, published, and open to scrutiny — including setbacks. The say–do gap destroys trust faster than it was built.

◆ INTERACTIVE

CASE STUDY

When an Ad Told Customers Not to Buy: Patagonia

“Don’t Buy This Jacket” — Black Friday, 25 November 2011

Patagonia ran a full-page New York Times ad for its best-selling R2 fleece, listing the jacket’s real environmental costs and directing readers to its Common Threads Initiative — repair, reuse, resale, recycling.

Why It Was Credible

It rested on real prior action: repair services, take-back, used-gear resale (later Worn Wear), and unusual openness about the impacts of its own products.

The Reported Result

Sales rose roughly 30% in the months that followed, as widely reported; the exact window and causation vary by source.

The Honest Reading

The ad built trust and preference; it did not obviously reduce total consumption. Imitators without Patagonia’s track record risk a backlash.

The Lesson

Credibility is earned by deeds first. The ad worked because the company had already earned the right to say it.

B2B, Employer, and Investor Audiences

Sustainable branding is not only aimed at shoppers — your most demanding audiences read the evidence closely.

Business Customers

Large buyers now ask suppliers for product carbon footprints and EPDs — recall Walmart's Project Gigaton reaching 5,000+ suppliers. Verified data is sales collateral.

Employer Brand

Sustainability performance influences recruitment and retention, especially among graduates — the audience in this room.

Investors and Regulators

Marketing claims and disclosures such as BRSR are read side by side; inconsistency between them is itself a red flag.

SECTION 4 · MEASUREMENT

How to Tell Whether Sustainable Marketing Is Working

Both commercial results and integrity need KPIs — and both should be reported.



Revenue Share

Share of revenue from offerings meeting a published sustainability definition — define the criteria first.



Claim Coverage

Percentage of public claims with an evidence file and legal review before publication.



Trust & Complaints

Brand-trust scores, greenwashing complaints, and regulator queries — the early-warning indicators.



Behaviour Outcomes

Repeat purchase, repair and take-back uptake, and refill rates — proof customers actually acted.

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

What made Patagonia's "Don't Buy This Jacket" credible rather than cynical?

A It was the cheapest ad it could buy

B It was backed by years of repair, reuse, and transparency actions ✓

C It guaranteed that sales would fall

D It avoided mentioning the jacket's environmental impact

ANSWER & WHY

B. It was backed by years of repair, reuse, and transparency actions

The ad's credibility came from prior action — repair services, resale, and honest impact disclosure. Without that track record, the same message would look like greenwashing with a twist.

BRINGING IT TOGETHER

Marketing Can Only Promise What Other Functions Deliver

Integrating sustainability into business functions means each function supplies the proof that marketing then communicates.

R&D / Design	LCA-informed product improvements that make real, specific claims possible.
Procurement	Certified inputs and supplier data (Scope 3 Category 1) to back sourcing claims.
Operations & Logistics	Measured energy, waste, and transport performance, including mode-shift gains.
Finance	Investment for greener products, and numbers that match what is disclosed (for example BRSR).
Legal & Compliance	Claim review against CCPA- and EU-style rules before anything is published.
IT & Data	Traceability and data systems so every claim can be evidenced on demand.

BRINGING IT TOGETHER

The Sustainable Marketing Loop

Four steps, repeated: each one is only credible if the previous one is real.

1. Develop	Design offerings with measurably lower impact, guided by LCA and customer insight.
2. Substantiate	Build the evidence file: data, certifications, EPDs, and legal review.
3. Communicate	Apply the five principles: specific, substantiated, relevant, complete, proportionate.
4. Verify & Improve	Track KPIs and complaints, correct mistakes openly, and feed lessons back into design.

◆ INTERACTIVE

APPLY IT ALL

“Pick an advertisement you have seen recently. Which of the five principles does it meet — and which would a regulator challenge first?”

This is exactly how a claims-review meeting begins in practice.

⏱ 4 MIN

B.TECH FINAL YEAR · COMPUTER SCIENCE & ENGINEERING · OPEN ELECTIVE

INTEGRATING SUSTAINABILITY INTO BUSINESS FUNCTIONS

Sustainable Finance

Accessing green finance and integrating ESG factors into financial decision-making

COURSE

Sustainability & Climate Change

KEY FRAMEWORKS

ICMA Principles · SEBI BRSR · ISSB

FORMAT

Lecture + Worked Problems

Labelled sustainable bonds have passed US$6.8 trillion. Yet two ESG raters agree on a company's score barely half the time.

Money is flowing toward sustainability faster than the measurement behind it is maturing. Today you learn both sides: how to raise green capital, and how to tell whether the "ESG" on the label is real.

BY THE END OF THIS LECTURE

You will be able to...



Classify

Tell green, social, sustainability-linked and transition instruments apart, and say what each promises.



Calculate

Work out a greenium, a carbon-adjusted valuation, and a portfolio's carbon intensity (WACI).



Integrate

Show where ESG enters valuation, credit decisions and capital budgeting.



Evaluate

Spot greenwashing and rating divergence, and apply India's SEBI and BRSR framework.

TODAY'S JOURNEY

Finance: The Function That Prices Everything Else

After marketing, finance is the next function to integrate. It does two jobs for sustainability: it supplies capital to green projects, and it prices sustainability risk into every decision the other functions make.



01

Accessing Green Finance

The instruments, the rules, and how a company or government actually raises green money.



02

Integrating ESG into Decisions

Where ESG enters valuation, credit, and investment strategy, and why ratings disagree.



03

Measuring & Disclosing

BRSR, ISSB, and the metrics (WACI) that make financial claims checkable.



04

Integrity & Limits

Greenwashing in finance, the DWS case, and what the evidence does and does not show.

SECTION 1 OF 4

Accessing Green Finance

Capital is the raw material of every sustainability project. Here is where it comes from.

SECTION 1 · DEFINITION

What Is Sustainable Finance?

Two directions of travel, often confused.

Financing Sustainability

Directing capital to projects and companies that deliver environmental or social benefit: solar parks, electric rail, efficient buildings, affordable housing.

Making Finance Sustainable

Taking environmental, social and governance (ESG) factors into every financial decision (lending, investing, valuing, insuring) because they change risk and return.

Two Lenses on Materiality

Financial materiality asks how sustainability affects the company's value. Double materiality adds how the company affects people and the planet. ISSB standards focus on the first; EU rules use the second.

A Map of Sustainable Debt Instruments

Most labelled sustainable finance is debt. What differs is what the label actually promises.



Green Bonds

Proceeds are earmarked for eligible green projects. Repayment is the issuer's ordinary obligation.



Social & Sustainability Bonds

Proceeds fund social projects (social bonds) or a mix of green and social projects (sustainability bonds).



Sustainability-Linked Bonds

Money can be used for anything, but the cost of the bond changes if the issuer misses a stated sustainability target.



Green Loans & SLLs

The bank-loan versions: use-of-proceeds green loans, and sustainability-linked loans whose margin moves with KPIs.



Transition Finance

Funding for high-emitting sectors to decarbonise credibly, such as steel, cement, and power. Controversial, and only as credible as the transition plan.



Blended Finance

Public or philanthropic money absorbs part of the risk to attract private capital into projects that would not otherwise be bankable.

Use-of-Proceeds vs. KPI-Linked: What Is Actually Promised?

Green Bond (Use of Proceeds)

What the money is for: specific, eligible green projects.

What is promised: proceeds are tracked and reported against the projects.

Pricing consequence: usually none after issue; the label can bring a small pricing benefit (“greenium”).

Main integrity risk: projects that are not truly green, or proceeds that are not tracked.

Sustainability-Linked Bond (KPI)

What the money is for: general corporate purposes. Nothing is ring-fenced.

What is promised: the issuer will hit a company-level target (the SPT) by a date.

Pricing consequence: the coupon steps up (typically 25 bp) if the target is missed.

Main integrity risk: targets that are easy to meet, or a step-up too small to matter.

SECTION 1 · THE MARKET STANDARD

The Four Core Components of the ICMA Green Bond Principles

Voluntary process guidelines, updated June 2025, that the market treats as the de facto standard.



1. Use of Proceeds

Proceeds go to eligible green projects with clear environmental benefits, described in the bond's legal documentation.



2. Project Selection

The issuer explains how projects are evaluated, which environmental objectives they serve, and how risks are managed.



3. Management of Proceeds

Proceeds are tracked, for example in a sub-account or sub-portfolio, and the balance is attested by the issuer.



4. Reporting

At least annual reporting on the allocation of proceeds and, where feasible, on expected environmental impact.

How Big Is Labelled Sustainable Debt?

US$6.8 Tn

Cumulative “aligned” green, social, sustainability and sustainability-linked bonds by end-2025 (Climate Bonds Initiative)

64%

Share of 2025 aligned issuance that carried the green label; green passed US$4 trillion cumulative

45%

Europe’s share of annual 2025 aligned issuance. Development banks hold a similar share of the cumulative market

2.2%

Share of all bond issuance that labelled bonds represented in 2024, a record of over US$1 trillion (IFC–Amundi)

Large in absolute terms, small as a share of the global bond market. The opportunity (and the measurement challenge) is the other 97–98%.

Sources: Climate Bonds Initiative, Global State of the Market 2025 (March 2026) and press release (Sept 2025); IFC–Amundi, Emerging Market Green Bonds 2024.

◆ INTERACTIVE

CASE STUDY

A Sovereign Goes Green: India's Sovereign Green Bonds



Government of India Sovereign Green Bonds (SGrBs): framework 9 Nov 2022; first auctions Jan–Feb 2023

The Ministry of Finance issued a framework aligned to the ICMA Green Bond Principles, then raised ₹16,000 crore in two ₹8,000 crore auctions (25 January and 9 February 2023), split across five- and ten-year maturities.

The Price Signal

The five-year bond cleared at 7.10% against 7.16% for a comparable conventional G-Sec, and the ten-year at 7.29% against 7.35%: a greenium of about 6 basis points, above the 2–3 bp expected.

The Safeguards

Proceeds go into the Consolidated Fund of India and finance public-sector projects that cut the economy's carbon intensity; a Green Finance Working Committee, chaired by the Chief Economic Advisor, oversees selection.

The Honest Caveat

Early demand was strong, but a later auction sold only ₹1,502 crore of ₹5,000 crore on offer, with primary dealers taking the rest. A label alone does not guarantee liquidity.

The Lesson

A credible framework can earn a small pricing benefit. Sustained demand depends on size, liquidity, and the broader state of the bond market.

SECTION 1 · WORKED EXAMPLE

Pricing the Greenium on a Government Tranche

GIVEN

- Five-year tranche size: ₹4,000 crore (illustrative split of the January 2023 auction)
- Green bond yield: 7.10%; comparable conventional G-Sec yield: 7.16%
- Assume the yield difference holds for the full five-year term

FORMULA

Greenium = Yield (conventional) − Yield (green)
Annual Saving = Principal × Greenium

STEP-BY-STEP CALCULATION

1. Greenium: 7.16% − 7.10% = 0.06% = 6 basis points
2. Annual interest saving: ₹4,000 crore × 0.0006 = ₹2.4 crore per year
3. Over five years (undiscounted): ₹2.4 crore × 5 = ₹12 crore

≈ ₹12 crore

Small next to the principal, which is why most issuers pursue green labels for investor diversification and reputation as much as price. (Illustrative: actual coupons are set at auction.)

◆ INTERACTIVE

CASE STUDY

Putting Money Where the Targets Are: Enel's First SLB

Enel, September 2019: the first sustainability-linked bond

The Italian utility issued a US$1.5 billion five-year bond whose coupon depends on a company-wide KPI: raising renewable capacity to at least 55% of total installed capacity by the end of 2021, up from about 46% in mid-2019.

The Mechanism

If the target is missed, the coupon steps up by 25 basis points. The money itself is not ring-fenced for any particular project.

The Market Response

Enel said the deal priced roughly 10 bp inside where a conventional bond would have priced, according to a law-firm case note. Demand was strong, and the structure was widely copied.

The Standard That Followed

ICMA published the Sustainability-Linked Bond Principles in June 2020: material KPIs, ambitious and benchmarked targets, a defined financial consequence, annual reporting, and independent verification.

The Criticism

Critics ask whether a 25 bp step-up is large enough to change behaviour, and whether targets are ambitious or merely continue existing trends. Both are questions an investor must test.

Where Indian Issuers and Projects Find Green Capital



Sovereign Green Bonds

The government's own green borrowing, with a published framework and allocation to public-sector green projects such as rail and renewables.



SEBI Green Debt Securities Rules

Listed green debt follows SEBI's revised framework, aligned to ICMA principles, plus "dos and don'ts" guidance issued to prevent greenwashing.



Public Green Lenders

Institutions such as IREDA lend specifically to renewable energy and efficiency projects, often at tenors commercial banks avoid.



Development Banks & Climate Funds

Multilateral lenders provide long-tenor and blended finance; development banks now hold roughly 45% of the cumulative aligned labelled debt market.

SECTION 1 · THE PROCESS

How a Company Raises Green Finance: Five Steps

The sequence mirrors the four ICMA components, with external review added before the bond is sold.

1. Build the Framework	Define eligible projects, selection process, proceeds management, and reporting in a Green Bond Framework.
2. Get It Reviewed	Obtain a Second Party Opinion or other external review confirming alignment with the principles.
3. Issue and Allocate	Sell the bond and allocate the proceeds to the eligible projects identified in the framework.
4. Track the Proceeds	Maintain a register so every rupee raised can be traced to a project, and any unallocated balance is held transparently.
5. Report	Publish annual allocation and impact reports, with figures (such as tCO_2e avoided) that can be checked.

◆ INTERACTIVE

CLASSIFY THE INSTRUMENT

Use-of-Proceeds, KPI-Linked, or At Risk?

For each description, decide which design it matches, or whether it is at risk as a sustainable label.

1. Proceeds restricted to metro-rail electrification, with annual allocation reports
2. Coupon rises 25 bp if emissions intensity misses the 2030 target; funds are for general purposes
3. Bond labelled “sustainable” with no framework, no tracking, and no reporting
4. Loan margin falls if the borrower meets a water-use reduction KPI
5. Government bond financing electric locomotives under an ICMA-aligned framework
6. Sustainability-linked bond whose target was already met when it was issued

USE OF PROCEEDS	KPI-LINKED	AT RISK
1	2	3
5	4	6

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

A bond's proceeds can be used for any purpose, but its coupon rises 25 bp if the issuer misses a carbon-intensity target. What is it?

A. A green use-of-proceeds bond

B. A sustainability-linked bond ✓

C. A social bond

D. An ordinary bond with no sustainability feature

ANSWER & WHY

B. A sustainability-linked bond

The financial consequence is tied to a company-level target (the SPT), and the proceeds are unrestricted. That is the defining design of a sustainability-linked bond, as in Enel's 2019 issue.

SECTION 2 OF 4

Integrating ESG into Decisions

Raising green money is half the job. The other half is making every ordinary decision see ESG risk.

ESG as Risk and Return Information

Investors use ESG less as an ethical statement and more as information about risk, in the same way they read leverage or governance.

Environmental

Emissions, energy and water use, pollution, and exposure to physical climate hazards and carbon pricing.

Social

Workforce safety and diversity, supply-chain labour standards, community relations, and product safety.

Governance

Board independence, executive pay, audit quality, and ethics. Often the most financially predictive pillar.

Seven Ways Investors Use ESG

The Global Sustainable Investment Alliance groups practice into seven strategies. Most funds combine several.



Exclusion Screening

Removing sectors or companies, such as coal or tobacco, from the investable universe.



Best-in-Class

Choosing the highest ESG performers within each sector, rather than excluding sectors.



Norms-Based Screening

Screening against international standards such as the UN Global Compact or OECD guidelines.



ESG Integration

Including ESG factors in financial analysis alongside traditional factors, such as in forecasts and valuation.



Thematic Investing

Targeting themes such as clean energy, water, or sustainable agriculture.



Impact Investing

Investing with the intention to generate measurable social or environmental benefit alongside a financial return.



Engagement & Voting

Using shareholdings to press management on ESG issues, rather than selling.

Three Channels Through Which ESG Reaches Value

A finance team does not need a separate “ESG model”. It needs ESG to show up in the models it already has.

Cash Flows	Carbon costs, energy savings, regulatory fines, lost customers, and new green revenues change forecast revenues and costs directly.
Discount Rate	Poor governance or high transition risk raises the return investors demand, lowering present value (as in the cost-of-capital example in our business-case lecture).
Terminal Value & Assets	Stranded-asset risk and the useful life of carbon-intensive plant change what a business is worth beyond the forecast period.
Capital Budgeting	Many firms apply an internal (shadow) carbon price to projects, so a rupee-per-tonne cost is visible before the investment is approved.

SECTION 2 · WORKED EXAMPLE

A Carbon-Adjusted Valuation

GIVEN

- Company EBITDA: ₹1,000 crore; valued at 8× EBITDA (EV = ₹8,000 crore)
- Scope 1+2 emissions: 1,000,000 tCO_2e per year
- Internal (shadow) carbon price used in analysis: ₹2,000 per tCO_2e

FORMULA

Carbon Cost = Emissions × Carbon Price
Adjusted EV = Multiple × (EBITDA − Carbon Cost)

STEP-BY-STEP CALCULATION

1. Carbon cost: 1,000,000 t × ₹2,000/t = ₹2,000,000,000 = ₹200 crore
2. Adjusted EBITDA: ₹1,000 crore − ₹200 crore = ₹800 crore
3. Adjusted EV: 8 × ₹800 crore = ₹6,400 crore (down from ₹8,000 crore)
4. Change in value: ₹1,600 crore, or 20% lower

−20% Value

A shadow carbon price turns an invisible risk into a number the investment committee can debate. (Illustrative: assumes no cost pass-through and no abatement.)

ESG in Credit: How Lenders Price Environmental Risk

Banks carry the long-dated exposures that carbon-intensive assets create, so ESG enters credit analysis as a risk factor.

Transition Risk	Loans to emitters can lose value if carbon prices, regulation, or demand shifts reduce future cash flows (the same logic as stranded assets).
Physical Risk	Flood, heat, and storm exposure can damage collateral such as factories and real estate, and interrupt customers' cash flows.
Project Finance: The Equator Principles	Launched in 2003 and based on the IFC Performance Standards, they commit adopting banks to assess and manage environmental and social risk in project loans.
Portfolio Steering	Sector limits, internal carbon prices, and transition-plan requirements for large emitters direct credit toward lower-risk clients.

ESG Ratings Do Not Agree With Each Other

0.54

Average correlation between the ESG ratings of six major providers (Berg, Kölbel & Rigobon, 2022)

0.38–0.71

The range across provider pairs; the best-agreeing pair reached 0.71

≈ 0.92–0.99

Correlation between credit ratings from Moody's and S&P, depending on how it is measured

The disagreement comes from three sources: scope (different attributes), measurement (different indicators for the same attribute, the largest source), and weights (different importance).

Source: Berg, Kölbel & Rigobon, "Aggregate Confusion: The Divergence of ESG Ratings," Review of Finance 26(6), 2022; MIT Sloan Aggregate Confusion Project.

Regulating the Raters

India chose to bring ESG raters under securities regulation rather than leave the market unsupervised.

ESG Rating Providers (ERPs)

In 2023 SEBI created a regulatory framework for ESG rating providers by amending its Credit Rating Agencies Regulations, 1999.

Data Behind the Ratings

SEBI's consultation proposed "Core ESG" ratings built on audited BRSR Core data, so raters work from assured numbers, not self-declared ones.

Why It Matters to You

An investment committee that relies on a rating inherits its methodology. Know what a rating measures and how it is weighted before using it.

◆ INTERACTIVE

THINK & SHARE

“Two ESG raters give your key supplier opposite scores. Which do you trust, and what exactly would you ask each rater?”

Think about scope, measurement, and weights, the three sources of divergence.

⏱ 3 MIN

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

Why do ESG ratings from different providers diverge so much more than credit ratings?

A ESG raters measure different attributes, use different indicators for the same attribute, and weigh them differently ✓

B ESG ratings are required by law to differ

C Credit ratings ignore risk entirely

D ESG raters only cover small companies

ANSWER & WHY

A. ESG raters measure different attributes, use different indicators for the same attribute, and weigh them differently

Berg, Kölbel and Rigobon decompose divergence into scope, measurement, and weights. Credit ratings share a single, well-defined target (default risk), so they agree far more.

SECTION 3 OF 4

Measuring & Disclosing

A claim that cannot be measured cannot be financed, priced, or audited.

Four Frameworks Behind the Numbers

Different regulators, converging content: governance, strategy, risk management, and metrics.



India: BRSR & BRSR Core

Mandatory for the top 1,000 listed companies since FY 2022–23; a focused BRSR Core subset needs external assessment or assurance on a phased timeline.



ISSB: IFRS S1 & S2

Global baseline standards issued in June 2023: general sustainability-related disclosures (S1) and climate-related disclosures (S2).



TCFD

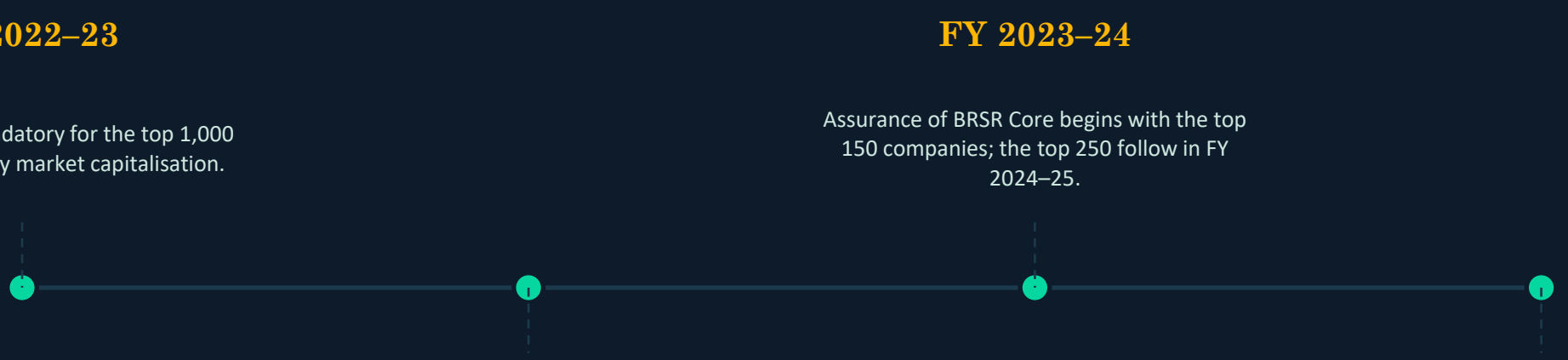
The Task Force on Climate-related Financial Disclosures set the four-pillar structure (governance, strategy, risk management, metrics and targets) that IFRS S2 builds on.



EU Taxonomy & SFDR

The Taxonomy classifies which activities count as environmentally sustainable; SFDR sets how funds disclose sustainability characteristics.

BRSR and BRSR Core: The Assurance Glide Path



What BRSR Core Asks Companies to Measure

Nine attributes, covering the environmental, social and governance questions investors ask most.

Environment	Greenhouse-gas footprint, water footprint, energy footprint, and waste management (embracing circularity).
Social	Employee wellbeing and safety, gender diversity, and inclusive development (such as sourcing from small and local producers).
Governance	Fairness in engaging with customers and suppliers, and openness of business (such as concentration of dealings with trading partners).

Financed Emissions: Scope 3 Category 15 in Practice

For a bank or fund, the biggest footprint is not its offices: it is the emissions of the companies it finances.

Investments Are Category 15

Recall from our footprint lecture that Scope 3 Category 15 covers investments. For banks and asset managers it dwarfs Scope 1 and 2.

The PCAF Standard

The Partnership for Carbon Accounting Financials provides the common method for calculating financed emissions across asset classes.

WACI: A Simple Portfolio Metric

Weighted average carbon intensity multiplies each holding’s weight by its emissions per unit of revenue. It is one of the metrics recommended by the TCFD.

SECTION 3 · WORKED EXAMPLE

Weighted Average Carbon Intensity (WACI) of a Portfolio

GIVEN

- Company A: 50% of portfolio; 100 tCO_2e per ₹ crore of revenue
- Company B: 30% of portfolio; 400 tCO_2e per ₹ crore of revenue
- Company C: 20% of portfolio; 50 tCO_2e per ₹ crore of revenue

FORMULA

WACI = Σ [Portfolio Weight × (Scope 1+2 Emissions ÷ Revenue)]

STEP-BY-STEP CALCULATION

1. A: 0.50 × 100 = 50; B: 0.30 × 400 = 120; C: 0.20 × 50 = 10
2. WACI = 50 + 120 + 10 = 180 tCO_2e per ₹ crore of revenue
3. Rebalance: move 10 percentage points from B to A (A 60%, B 20%, C 20%)
4. New WACI = 0.60 × 100 + 0.20 × 400 + 0.20 × 50 = 60 + 80 + 10 = 150

180 → 150 (−17%)

Shifting weight away from the most carbon-intensive holding cuts the metric. Caveat: intensity can fall just because revenue rises, so pair it with absolute emissions and real engagement.

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

In the WACI example, moving 10 percentage points of the portfolio from Company B (400) to Company A (100) changes WACI by how much?

A It rises by 30

B It falls by 30, from 180 to 150 ✓

C It falls by 10

D It stays the same

ANSWER & WHY

B. It falls by 30, from 180 to 150

Each point of weight moved from B to A removes (400 − 100) = 300 tCO_2e per ₹ crore × 0.10 = 30 from the weighted sum: 180 − 30 = 150.

SECTION 4 OF 4

Integrity & Limits

The same label that attracts capital also attracts scrutiny.

◆ INTERACTIVE

CASE STUDY

When ESG “In the DNA” Was Not in the Process: DWS

DWS Group, the asset-management arm of Deutsche Bank (2021–2025)

A former head of sustainability alleged that DWS overstated how far ESG was integrated into its investment process. The US SEC settled with DWS’s US adviser in September 2023, and German prosecutors followed.

The SEC Finding

On 25 September 2023 the SEC announced a US$19 million penalty, then its largest for ESG greenwashing by an asset manager, finding that staff did not follow the ESG process DWS had marketed. A separate US$6 million AML penalty was settled the same day.

The Trigger

The inquiry began after a whistleblower complaint from DWS’s former head of ESG, Desirée Fixler, in 2021. Police raided DWS and Deutsche Bank offices in Frankfurt in May 2022.

The Second Penalty

In 2025 the Frankfurt prosecutor’s office fined DWS €25 million over misleading statements about the ESG credentials of its products.

The Balance

DWS said the SEC order found no misstatements in its financial disclosures or fund prospectuses and that its marketing had sometimes been “exuberant”. The SEC’s finding concerned process versus promise.

SECTION 4 · WHERE INTEGRITY FAILS

Four Ways Sustainable Finance Overpromises

The same five-principle test from marketing applies: specific, substantiated, relevant, complete, proportionate.



Label Drift

Funds and bonds carry sustainable names, but the process behind the label is thin or not followed (the DWS pattern).



Soft Targets

Sustainability-linked targets that are easy to hit, already met, or backed by a step-up too small to change behaviour.



Untracked Proceeds

Green bond money that cannot be traced to eligible projects, or is never reported on.



Opaque Scores

ESG ratings or metrics without a published methodology, so users cannot tell what they are buying.

Does ESG Pay? An Honest Reading

The strongest claims on both sides go beyond what the research supports.

Mixed, Not Magic

Studies find ESG factors often act as risk indicators, but results vary by period, region, and method. “ESG always outperforms” is not established.

The Rater Problem Again

Because ratings diverge, a result built on one provider’s ratings might not replicate with another’s (Berg et al., 2022). Conclusions are only as good as the data behind them.

Small Price Benefits

The sovereign greenium in India was about 6 bp. Green labels help diversify investors but are not a large cost saving.

A Contested Label

In some markets “ESG” has become politically contested, and firms now talk more about risk, resilience, and transition. The underlying analysis (carbon cost, physical risk, governance) remains standard finance.

SECTION 4 · A CHECKLIST

Five Checks for Credible Sustainable Finance

Before you buy, lend, issue, or recommend, ask whether the label survives these tests.

1. Is the Use Defined?	Eligible projects or KPIs are specific, relevant to the issuer's real footprint, and written down.
2. Is It Reviewed?	An independent party has checked the framework or targets (Second Party Opinion, assurance).
3. Is the Target Ambitious?	Targets go beyond business-as-usual trends and can be benchmarked against peers or science.
4. Is Reporting Verified?	Allocation and impact are reported at least annually, with figures that auditors can test.
5. Is There a Consequence?	Missing a target changes price or terms in a way that matters, such as a meaningful step-up.

BRINGING IT TOGETHER

Finance Connects Every Business Function

Integrating sustainability across functions means each one both asks something of finance and gives it data to price.

R&D / Design	Needs capital for greener products; gives finance LCA data to size benefits.
Procurement	Needs working-capital terms for certified inputs; gives supplier emissions data (Scope 3 Cat. 1).
Operations	Needs capex for efficiency and renewables; gives measured energy, water, and waste data.
Marketing	Needs claims it can defend; finance's verified metrics are the evidence file.
Legal & Compliance	Reviews green bond and disclosure documents against SEBI and EU rules.
IT & Data	Builds the pipelines behind BRSR Core, financed-emissions, and WACI numbers.

◆ INTERACTIVE

APPLY IT ALL

“You are CFO of a mid-size manufacturer that needs ₹500 crore for a solar and efficiency retrofit. Which instrument would you choose, and what evidence file would you build?”

Consider use-of-proceeds vs. KPI-linked, external review, and the five integrity checks.

⏱ 4 MIN

INTEGRATING SUSTAINABILITY INTO BUSINESS FUNCTIONS

Innovation & Sustainability

Developing new sustainable technologies and business models

Solar electricity got about 90% cheaper in fourteen years. Yet the IEA still expects roughly a third of 2050's emission cuts to come from technology that is not yet on the market.

Innovation is both our best record and our largest unknown. Today covers how new technologies are built, financed and scaled, how business models decide whether they spread, and how to innovate without creating new problems.

TODAY'S JOURNEY

R&D: The Function That Creates What Does Not Yet Exist

After marketing and finance, innovation completes the picture: marketing communicates, finance funds, and innovation decides what there is to communicate and fund. It has two halves: technology and business model.



01

Why Innovation Is Central

What innovation is, how much of the climate solution depends on it, and what it has already delivered.



02

From Lab to Market

Readiness levels, the valley of death, green hydrogen, and AI-accelerated discovery.



03

Sustainable Business Models

Eight archetypes, the layered canvas, product-as-a-service, and the Better Place failure.



04

Innovating Responsibly

Rebound effects, forecasting humility, and green software (SCI) for your own field.

SECTION 1 OF 4

Why Innovation Is Central

Deploying what exists is not enough. Some of the solution has to be invented.

SECTION 1 · DEFINITION

What Is Sustainability-Oriented Innovation?

More than a greener version of the same product.

The Definition

New or improved products, processes, practices or business models that deliver environmental or social benefit as well as commercial value.

Efficiency Is Not Enough

Eco-efficiency improves what exists. Step changes need radical innovations (new materials, new energy carriers) and system change (new grids, new mobility networks).

How Clean Technologies Spread

Christensen’s disruption pattern fits many clean technologies: they begin in niches (off-grid power, early adopters), improve and get cheaper, then move into the mainstream.

Four Levels of Sustainability Innovation

Each level changes more of the system, and carries more risk and more potential.



Product

A better artefact: a higher-efficiency solar module, bio-based packaging, a longer-lasting battery.



Process

A better way to make it: hydrogen-based ironmaking instead of coal, or manufacturing with recycled feedstock.



Business Model

A better way to earn from it: leasing, refill, take-back, and pay-per-use instead of one-off sale.



System

A better arrangement around it: smart grids, integrated public transport, circular supply networks.

How Much of the Solution Is Not Yet on the Market?

≈ 46%

Share of 2050 emission cuts that the IEA's 2021 net-zero pathway drew from technologies not yet on the market ("almost half")

≈ 35%

The same share in the IEA's 2023 update, after technology-readiness upgrades and record growth of key clean technologies

> 80%

Share of reductions needed by 2030 delivered by proven, often cost-effective technologies in the 2023 pathway

Deploy what exists this decade; invent what does not for the next. Both are required, and they need different policies and different money.

Source: IEA, Net Zero by 2050 (2021) and Net Zero Roadmap update (September 2023), with IEA commentary on technology readiness.

The Cost Curves That Changed the Debate

−90%

Utility-scale solar PV electricity cost, 2010–2024: US$0.417 to US$0.043 per kWh (IRENA)

−87%

Total installed cost of utility-scale solar PV: US$5,283 to US$691 per kW

−93%

Battery storage installed cost, 2010–2024: US$2,571 to US$192 per kWh

US$0.038

India's utility-scale solar PV cost per kWh in 2024, below the global average of US$0.043

IRENA's next report puts solar at about US$44/MWh (89% below 2010) and a four-hour utility battery near US$140/kWh, roughly 95% below 2010.

Source: IRENA, Renewable Power Generation Costs in 2024 (July 2025) and 2025 (executive summary, 2026).

Learning Curves: Why Deployment Is Itself Innovation

The pattern behind the solar and battery numbers.

Wright's Law	Unit cost falls by a roughly constant percentage each time cumulative production doubles. T.P. Wright first described it for aircraft manufacturing in 1936.
Why It Happens	Scale economies, learning by doing, design simplification, supply-chain standardisation, and R&D that rides along with volume.
The Policy Lesson	Volume drives cost, so demand-pull (auctions, mandates, public procurement) is innovation policy, not only subsidy.
The Caveat	Learning rates are estimated from data, differ by technology, and are not guaranteed. Technologies limited by materials or physics may flatten.

SECTION 1 · WORKED EXAMPLE

A Learning Curve: Five Doublings of Production

GIVEN

- Initial unit cost: ₹100
- Learning rate: 20% cost reduction per doubling of cumulative production (illustrative)
- Cumulative production doubles five times

FORMULA

Cost after n doublings = $C_0 \times (1 - \text{learning rate})^n$

STEP-BY-STEP CALCULATION

1. Learning factor per doubling: 1 − 0.20 = 0.80
2. After five doublings: ₹100 × 0.80^5 = ₹100 × 0.32768 ≈ ₹32.8
3. Cumulative volume needed: 2^5 = 32 times the starting volume
4. Doublings to halve the cost: ln(0.5) ÷ ln(0.8) ≈ 3.1

−67% Cost

A cost cut of two-thirds needs a 32-fold rise in cumulative volume. That is why early demand matters so much. (The 20% rate is illustrative; real rates must be measured.)

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

With a 20% learning rate, how many doublings of cumulative production cut unit cost to about one-third of its starting level?

A Two doublings

B Five doublings ✓

C Ten doublings

D Twenty doublings

ANSWER & WHY

B. Five doublings

$0.80^5 \approx 0.33$. Five doublings means 32 times the cumulative volume, which is why demand-pull policies that create volume can be as important as R&D funding.

SECTION 2 OF 4

From Lab to Market

Most promising technologies die between the laboratory and the first commercial plant.

SECTION 2 · THE LADDER

Technology Readiness Levels (TRL): Nine Rungs from Idea to Market

Developed at NASA and now used by many funders, including the EU's Horizon programmes, to say how mature a technology is.

TRL 1–3 · Research	Basic principles observed, concept formulated, and experimental proof of concept in a laboratory.
TRL 4–6 · Development	Validated in the lab (4), validated in a relevant environment (5), and a prototype demonstrated in a relevant environment (6).
TRL 7–9 · Deployment	Prototype in an operational environment (7), system complete and qualified (8), and proven in real operation (9).
The Valley of Death	Roughly TRL 5–8: costs jump, research grants stop, and commercial investors do not yet see proven returns.

Why the Valley of Death Exists

The same gap appears in solar, batteries, hydrogen, and carbon capture.

Technology Risk and Market Risk Together	A first-of-a-kind plant must prove that it works and that customers will pay. Investors can price one of those risks, not easily both.
Capital Intensity	Moving from a ₹1 crore lab to a ₹200 crore demonstration plant multiplies the cheque while the evidence is still thin.
Who Bridges It	Grants and demonstration funds, public procurement and offtake guarantees, development banks, and blended finance (see our sustainable finance lecture).

SECTION 2 · WORKED EXAMPLE

The Innovation Funnel: Why Most Money Is Spent at the Last Gate

GIVEN

- Stage A (lab to prototype): ₹1 crore each; 50% succeed. Stage B (prototype to demo): ₹20 crore each; 40% succeed
- Stage C (demo to first commercial plant): ₹200 crore each; 70% succeed
- Start with a portfolio of 100 technologies (illustrative figures)

FORMULA

Survivors = Start × p(A) × p(B) × p(C)
Spend = Σ (technologies entering stage × cost per stage)

STEP-BY-STEP CALCULATION

1. Survivors: 100 × 0.5 × 0.4 × 0.7 = 14 technologies (14%)
2. Spend: A = 100 × ₹1 = ₹100 crore; B = 50 × ₹20 = ₹1,000 crore; C = 20 × ₹200 = ₹4,000 crore; total ₹5,100 crore
3. Cost per commercial success: ₹5,100 crore ÷ 14 ≈ ₹364 crore
4. Share of all spending at the last gate: ₹4,000 ÷ ₹5,100 ≈ 78%

78% at Gate C

Most money is spent where risk is still high and revenue not yet proven, which is why public capital and offtake commitments are decisive. Portfolio thinking matters: expect most to fail.

◆ INTERACTIVE

CASE STUDY

A Mission to Cross the Valley: India's National Green Hydrogen Mission



National Green Hydrogen Mission (NGHM): approved by the Union Cabinet on 4 January 2023

The Mission allocates ₹19,744 crore to build 5 million tonnes a year of green hydrogen capacity by 2030, with about 125 GW of new renewable capacity. ₹17,490 crore goes to the SIGHT incentive programme; the rest to pilots, R&D, and other components.

Supply-Push and Demand-Pull

SIGHT pays incentives for domestic electrolyser manufacturing and for green hydrogen production, while green ammonia purchase agreements anchor demand at fertiliser plants.

Awards So Far

By August 2026: 3,000 MW a year of electrolyser manufacturing awarded to 15 companies; 756,100 tonnes a year of hydrogen production capacity awarded; 670,000 tonnes a year of green ammonia contracted with 11 fertiliser units.

The Honest Gap

As of 6 August 2026 only ₹292 crore (about 1.5% of the outlay) had been spent, and awarded production capacity is about 15% of the 5 million tonne target.

The Lesson

Awards are not operating plants. Cost parity depends on electrolyser and renewable-power cost curves, and on demand that actually arrives.

◆ INTERACTIVE

CASE STUDY

AI as an Accelerator at the Front of the Pipeline: GNoME

Google DeepMind's GNoME: Graph Networks for Materials Exploration (Nature, November 2023)

A graph neural network trained with active learning predicted the stability of inorganic crystals and identified about 2.2 million candidate structures, of which roughly 380,000 are predicted most stable: a pre-filtered list for battery, solar and other clean-technology research.

The Scale

The Nature paper reports 381,000 newly discovered stable materials, almost an order of magnitude more than previous work, with predictions added to the open Materials Project database.

Validation So Far

736 of the structures have been independently made in laboratories worldwide, and an AI-guided autonomous lab at Berkeley synthesised 41 new materials in a companion study.

Prediction Is Not Product

Predicting stability is not synthesising a material, nor showing that it is useful, affordable, and manufacturable. Those steps sit at TRL 3–7 and cost most.

Treat With Care

Published critiques question how many of the "new" materials are truly novel and useful. Read the counts as candidates, not as discoveries.

◆ INTERACTIVE

PLACE IT ON THE LADDER

Research, Demonstration, or Commercial?

For each situation, decide which stage of the readiness ladder it describes.

1	A new electrolyte chemistry works in a laboratory coin cell
2	A 10 MW pilot plant runs in a real industrial setting for the first time
3	The same technology is replicated at several sites, selling output under long-term contracts
4	Basic principles are observed and reported in a journal paper
5	A prototype is tested in a relevant environment outside the lab
6	A complete, qualified system is proven in everyday operation

RESEARCH (1–3)	DEMO (4–7)	COMMERCIAL (8–9)
1	2	3
4	5	6

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

In the funnel example, why is public money or an offtake commitment most valuable at the last gate?

A The last gate is the cheapest stage

B It holds about 78% of the spend while commercial returns are still unproven ✓

C Research grants already cover it fully

D Banks always lend freely at that stage

ANSWER & WHY

B. It holds about 78% of the spend while commercial returns are still unproven

Stage C costs ₹4,000 crore of the ₹5,100 crore total, yet revenue is not proven until plants operate. Guaranteed demand and public capital reduce that risk enough for private money to follow.

SECTION 3 OF 4

Sustainable Business Models

A technology only matters if someone can profitably deliver it to many people.

Bocken's Eight Sustainable Business Model Archetypes

Grouped by Bocken, Short, Rana and Evans (2014) into technological, social, and organisational innovation.



Maximise Efficiency

Technological: do more with less material and energy, through lightweighting and lean design.



Value from "Waste"

Technological: treat one process's residue as another's input, as in industrial symbiosis.



Substitute with Renewables

Technological: replace fossil inputs with renewable energy and natural processes.



Functionality, Not Ownership

Social: sell the service or outcome and keep the asset, as in leasing and pay-per-use.



Stewardship

Social: take responsibility for the long-run welfare of stakeholders and resources.



Sufficiency

Social: encourage lower consumption through durability, repair, and moderation.



Re-purpose for Society

Organisational: put social or environmental mission at the core of the company.



Scale-up Solutions

Organisational: replicate impact quickly through partnerships, franchising, and open designs.

Each Archetype Has Real-World Examples

Illustrations of how the archetypes show up in different industries.

Maximise Efficiency	Lightweight packaging, lean manufacturing, and energy-efficient product design.
Value from "Waste"	Industrial symbiosis, such as Kalundborg in Denmark, where firms exchange steam, water, and by-products.
Substitute with Renewables	Solar PV replacing diesel generators; bio-based materials replacing fossil-based ones.
Functionality, Not Ownership	Lighting-as-a-service, mobility subscriptions, and battery leasing.
Stewardship	Certified supply chains such as Fairtrade and FSC; take-back and responsible sourcing.
Sufficiency	Durable, repairable products and repair services, such as Patagonia's Worn Wear programme.
Re-purpose for Society	Social enterprises, including pay-as-you-go solar for off-grid households.
Scale-up Solutions	Open-source designs, franchising, and partnerships that let others replicate quickly.

SECTION 3 · A DESIGN TOOL

The Layered Business Model Canvas

The Business Model Canvas (Osterwalder and Pigneur) describes how a company creates and captures value. Joyce and Paquin (2016) extended it to three layers.

Economic Layer	The familiar canvas: value proposition, customers, channels, revenue streams, cost structure, key partners, resources, and activities.
Environmental Layer	A life-cycle view of the same model (materials, production, distribution, use, end-of-life) showing environmental impacts and benefits. In effect, LCA on the canvas.
Social Layer	A stakeholder view: who is affected, and what social impacts and benefits (jobs, access, health, community) the model creates or removes.
Why Use It	It puts value, hidden costs, and affected people on one page, so a team can test whether profit relies on impacts someone else pays for.

Selling a Product vs. Selling an Outcome

Selling a Product

Revenue arrives once, at the sale. Profit rises with units sold.

The seller has little reason to design for long life or easy repair; durability can reduce future sales.

The customer carries performance, maintenance, and end-of-life risk.

Barriers are modest: familiar to customers, finance teams, and accountants.

Product-as-a-Service

Revenue arrives over time, per use or per period. Profit rises with uptime and long life.

The seller keeps ownership, so durable, repairable, upgradeable design lowers its own costs.

The provider carries those risks and needs data to manage them: a natural role for IoT and software.

Barriers: capital tied up in assets, customer trust, new contracts, and pricing for use.

◆ INTERACTIVE

CASE STUDY

When a Systems Innovation Fails to Scale: Better Place



Better Place (2007–2013): swappable batteries as a service

Better Place sold electric cars without batteries: drivers leased the battery and swapped it for a fresh one at automated stations in minutes. It raised about US$850 million from investors including HSBC, Morgan Stanley, and GE, and went into liquidation in 2013.

The Idea

Separate car from battery to cut upfront cost and remove range anxiety. It was product-as-a-service logic applied to the most expensive component.

The Numbers

Well under 2,000 cars were deployed across Israel and Denmark by most accounts, and only a few dozen swap stations were built, against an early forecast of 100,000 cars on Israeli roads by 2010.

Why It Failed

Heavy infrastructure spending before revenue; one car model from one manufacturing partner; leasing firms and drivers unconvinced of the economics; permit and rollout delays.

The Lesson

A sound idea can fail on ecosystem coordination, capital intensity, and timing. Swapping is now being retried mainly for fleets and small vehicles, where standard batteries and predictable routes help.

◆ INTERACTIVE

MATCH THE ARCHETYPE GROUP

Technological, Social, or Organisational?

For each example, decide which group of Bocken's archetypes it belongs to.

1. Lightweighting a product to cut material and shipping energy
2. Selling lighting as a service while keeping ownership of the fixtures
3. Using one factory's waste heat as another factory's input
4. Encouraging repair and long use instead of frequent replacement
5. Open-sourcing a design so others can replicate it widely
6. Converting a company into a social enterprise with a mission at its core

TECHNOLOGICAL	SOCIAL	ORGANISATIONAL
1	2	5
3	4	6

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

A provider installs lighting, keeps ownership, and charges customers per unit of light delivered. Which archetype is this?

A Encourage sufficiency

B Create value from waste

C Deliver functionality, rather than ownership ✓

D Develop scale-up solutions

ANSWER & WHY

C. Deliver functionality, rather than ownership

Selling an outcome while retaining the asset is the “functionality rather than ownership” archetype. It aligns the provider’s profit with durable, efficient, repairable design.

SECTION 4 OF 4

Innovating Responsibly

The best innovation solves the problem without moving it somewhere else.

SECTION 4 · UNINTENDED CONSEQUENCES

Rebound, Problem-Shifting, and Responsible Innovation

Innovation can backfire in predictable ways, which means it can be managed.

Rebound Effects	Efficiency gains can be partly or fully offset by extra use (the Jevons paradox from our principles lecture), so innovation needs caps or price signals beside it.
Problem-Shifting	Solving one problem can create another: battery minerals and their supply chains, panel and battery end-of-life, and the electricity demand of data centres. An LCA lens catches these early.
Four Habits of Responsible Innovation	Stilgoe, Owen and Macnaghten (2013): anticipate impacts, reflect on assumptions, include affected people, and respond by changing course.
Who Gets the Benefit?	Technologies priced for rich markets may not reach communities most exposed to climate impacts. Social equity applies to innovation too.

Even the IEA's Technology Forecasts Moved

Between the 2021 and 2023 editions of its net-zero roadmap, some technologies surged and others stalled.

What Slipped

Solid-state battery prototypes saw production delays; fuel-cell electric vehicles have not reached the market potential once foreseen; carbon capture on power plants is moving more slowly than projected.

What Surged

Key clean technologies saw record growth, and the share of 2050 reductions from not-yet-marketed technologies fell from almost half to around 35%.

The Planning Lesson

Think in portfolios: back many options, expect most to miss (our funnel example: 14 of 100 survive), and be ready to scale winners fast.

SECTION 4 · YOUR OWN FIELD

Green Software: The Software Carbon Intensity (SCI) Standard

For computer scientists, innovation for sustainability includes making software itself lower-carbon.

What It Is	ISO/IEC 21031:2024, published March 2024, from the Green Software Foundation: a method to calculate the carbon emission rate of a software system.
The Formula	SCI = ((E × I) + M) per R. E = energy used (kWh); I = carbon intensity of the grid where and when it runs (gCO_2e/kWh); M = embodied hardware emissions; R = functional unit (per user, request, or transaction).
Why “Per R” Matters	R is the LCA functional unit again. A rate lets you compare designs and track progress as usage grows: totals can rise while the rate falls.
What It Excludes	Offsets and credits cannot lower an SCI score. Only real reductions in E, I, or M count, consistent with our view of offset quality.

SECTION 4 · WORKED EXAMPLE

Calculating and Lowering a Software Carbon Intensity Score

GIVEN

- An API handles R = 1,000 requests using E = 0.5 kWh of server energy
- Grid carbon intensity where it runs: I = 700 gCO_2e/kWh
- Embodied hardware emissions attributed to this workload: M = 20 gCO_2e (illustrative)

FORMULA

SCI = ((E × I) + M) per R

STEP-BY-STEP CALCULATION

1. Operational part: 0.5 kWh × 700 gCO_2e/kWh = 350 gCO_2e
2. SCI = 350 + 20 = 370 gCO_2e per 1,000 requests
3. Option 1, move to a cleaner region (I = 100): (0.5 × 100) + 20 = 70, a fall of 81%
4. Option 2, halve the energy by optimising code (E = 0.25): (0.25 × 700) + 20 = 195, a fall of 47%

370 → 70 or 195

Both options beat buying offsets, which cannot lower SCI. Which is better depends on grid mix and effort; the best designs do both. (All figures are illustrative.)

◆ INTERACTIVE

THINK & SHARE

“Pick one: green hydrogen, battery swapping, or AI-designed materials. Where does it sit on the readiness ladder, which business model would get it adopted, and what could go wrong?”

Use the four lenses from today: readiness, funding gate, archetype, and unintended consequences.

⏱ 4 MIN

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

In SCI = ((E × I) + M) per R, which action can NOT lower the score?

A Running the workload where the grid is cleaner

B Optimising code so it uses less energy

C Purchasing carbon offsets ✓

D Extending the life of the hardware that runs the workload

ANSWER & WHY

C. Purchasing carbon offsets

SCI counts real emissions: energy (E), grid carbon intensity (I), and embodied hardware emissions (M). Offsets and credits are explicitly not a way to reduce an SCI score.

BRINGING IT TOGETHER

Five Questions for Any Sustainable Innovation

Use these before backing, building, or buying a new technology or business model.

1. Does It Hit a Hotspot?	Does it address a material impact found by LCA or Scope 3 analysis, rather than only an easy one?
2. Where Is It on the Ladder?	What is its TRL, and who funds the next gate (grant, blended finance, offtake)?
3. What Model Gets It Adopted?	Which archetype aligns the provider’s profit with durability and customer value?
4. What Could Go Wrong?	Rebound, problem-shifting, supply-chain risk, end-of-life, and who is left out.
5. Can We Prove It?	Is there an evidence file (LCA, EPD, SCI score) that survives CCPA- and EU-style claim rules?

BRINGING IT TOGETHER

Innovation Connects Every Business Function

The functions covered in this series each supply something innovation needs.

R&D / Design	Runs the TRL pipeline and builds LCA into every stage gate.
Finance	Bridges the valley of death with grants, blended finance, and green bonds for first-of-a-kind assets.
Procurement	Secures critical materials responsibly and tests supplier innovations.
Operations	Hosts pilots and scales what works.
Marketing	Tests real demand and communicates claims credibly.
Legal & IT	Protect IP, manage regulation, and build the data pipelines (SCI, LCA) that prove impact.

◆ INTERACTIVE

APPLY IT ALL

“You lead innovation at a mid-size Indian manufacturer. Pick one product or process and pitch in three minutes: the hotspot, the readiness level, the business model, and the biggest risk.”

This is how a real sustainability innovation review begins.

⏱ 4 MIN

Measuring & Reporting Sustainability Performance

Key metrics, frameworks, and reporting standards

The rulebook for sustainability reporting is being rewritten as we speak.
The discipline underneath it is not.

ISSB standards adopted by 28 jurisdictions, the EU cutting mandatory datapoints by about 60%, India's assurance net widening to the top 1,000 companies, and new GRI climate rules from 2027. Whatever the framework: define the metric, set the boundary, prove the number.

TODAY'S JOURNEY

From a Number to a Credible Disclosure

Earlier lectures taught you to calculate footprints and read BRSR and ISSB at a glance. Today covers the full chain: which metrics to track, which framework to report under, how the data becomes a disclosure, and how it earns trust.



01

Choosing the Right Metrics

What makes a good metric, absolute versus intensity, and targets that follow the science.



02

The Framework Landscape

GRI, ISSB, TCFD, CDP, GHG Protocol, SBTi, ESRS and BRSR, and how they converge.



03

From Data to Disclosure

Materiality, boundaries, base years, data quality, and the ESG data pipeline.



04

Credibility & Assurance

Limited and reasonable assurance, ISSA 5000, warning signs, and the road ahead.

SECTION 1 OF 4

Choosing the Right Metrics

Everything downstream (frameworks, assurance, targets) depends on measuring the right thing.

SECTION 1 · THE STANDARD FOR A METRIC

Five Tests of a Good Sustainability Metric

Run any proposed indicator through these before adding it to a dashboard or a report.



Relevant

Tied to a material impact or risk of this business, not a metric that is merely easy to collect.



Measurable

Defined with a unit, formula, and data source so two people get the same number.



Comparable

Consistent over time and across peers: same boundary, same method, same units.



Verifiable

Backed by evidence an independent party can test: meters, invoices, contracts.



Actionable

Moves when someone acts. A metric nobody can influence is reporting, not management.

Inputs, Outputs, Outcomes, and Leading vs. Lagging

A balanced scorecard mixes types, because each answers a different question.

Inputs	Resources and effort committed: renewable electricity purchased, training hours, R&D spend.
Outputs	Direct results of activity: kWh saved, tonnes of waste diverted, suppliers audited.
Outcomes	Real-world change: emissions avoided, injury rates falling, water stress reduced. The hardest to measure and the most meaningful.
Leading vs. Lagging	Leading indicators predict (near-misses reported, supplier audits completed); lagging indicators record what happened (lost-time injuries, annual emissions). Good dashboards pair both.

Five Families of Sustainability Metrics

Most frameworks draw from these families, which is why one dataset can serve many reports.

Climate & Energy	Scope 1, 2 and 3 emissions (tCO_2e); energy consumed (GJ or MWh) and renewable share; emissions intensity.
Water & Waste	Water withdrawal and consumption (m^3), including in water-stressed areas; waste generated and diversion rate.
Nature	Land use, sites near sensitive areas, and the share of deforestation-free sourcing.
People	Lost-time injury frequency rate (injuries × 1,000,000 ÷ hours worked); gender diversity; training; fair wages.
Governance	Board independence, ethics and anti-corruption training, supplier code compliance, data-privacy incidents.

Absolute vs. Intensity Metrics

Absolute Metric

Total emissions in tCO_2e, regardless of how much the company produces.

Answers: is our contribution to atmospheric carbon actually falling?

Used by science-based targets and carbon-budget logic.

Risk: can look like failure for a company that grows while cutting waste.

Intensity Metric

Emissions per unit of activity: per ₹ crore revenue, per tonne of product, per employee.

Answers: are we getting more efficient per unit of output?

Used for benchmarking, for growing firms, and in India's intensity-based CCTS targets.

Risk: can show progress while the total keeps rising (see next slide).

SECTION 1 · WORKED EXAMPLE

When Intensity Falls but Total Emissions Rise

GIVEN

- Year 1: 100,000 tCO_2e emitted on revenue of ₹1,000 crore
- Year 5: revenue grows 30% to ₹1,300 crore
- Year 5: emissions intensity (tCO_2e per ₹ crore) is 20% lower than Year 1

FORMULA

Absolute Emissions = Intensity × Activity

STEP-BY-STEP CALCULATION

1. Year 1 intensity: 100,000 ÷ 1,000 = 100 tCO_2e per ₹ crore
2. Year 5 intensity: 100 × (1 − 0.20) = 80 tCO_2e per ₹ crore
3. Year 5 absolute: 80 × 1,300 = 104,000 tCO_2e
4. Change in absolute emissions: (104,000 − 100,000) ÷ 100,000 = +4%

+4% Absolute

A 20% intensity improvement coincided with higher total emissions. Honest reports show both numbers, because they answer different questions. (Figures are illustrative.)

SECTION 1 · WORKED EXAMPLE

A Science-Based Glide Path: The 4.2% Rule

GIVEN

- Base year 2020 Scope 1+2 emissions: 100,000 tCO_2e
- SBTi minimum for a 1.5°C-aligned target: 4.2% of base-year emissions per year, linear
- Target year 2030 (ten years after the base year)

FORMULA

Target Emissions = Base – (4.2% × Base × Years)

STEP-BY-STEP CALCULATION

1. Annual reduction: 0.042 × 100,000 = 4,200 tCO_2e per year
2. Ten-year reduction: 4,200 × 10 = 42,000 tCO_2e (42%)
3. 2030 target: 100,000 − 42,000 = 58,000 tCO_2e
4. Mid-point check: the 2025 path value is 79,000; actual 85,000 would be 6,000 tCO_2e off track

58,000 t by 2030

The SBTi's minimum near-term ambition for Scope 1+2 is a 42% cut by 2030 from a 2020 base. Its April 2026 update adjusts annual rates for later base years while keeping the 4.2% floor.

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

A company cuts emissions intensity by 20% while revenue grows 30%. What happens to its absolute emissions?

A They fall by about 20%

B They rise by about 4% ✓

C They stay exactly the same

D They rise by 30%

ANSWER & WHY

B. They rise by about 4%

Absolute = intensity × activity, so 0.80 × 1.30 = 1.04. Intensity targets can hide rising totals, which is why science-based targets use absolute reductions.

SECTION 2 OF 4

The Framework Landscape

Many frameworks, different audiences, and a slow convergence toward one global baseline.

SECTION 2 · THE LANDSCAPE

Six Names You Will Meet

They are not interchangeable: some set what to disclose, some how to count, some how ambitious to be.



GRI

Global Reporting Initiative: impact-focused standards covering an organisation's effects on the economy, environment, and people. Voluntary, with universal standards since 2023.



ISSB (IFRS S1 & S2)

Investor-focused baseline for sustainability and climate disclosure, built on TCFD and SASB. Applies only where a jurisdiction adopts it.



TCFD

A four-pillar climate disclosure structure (governance, strategy, risk management, metrics and targets). Disbanded in 2023; monitoring passed to the IFRS Foundation.



CDP

A disclosure platform where companies answer standard climate, water and forest questionnaires requested by investors and buyers.



GHG Protocol & SBTi

The accounting rules for emissions (Scopes 1, 2, 3) and the initiative that validates science-based targets.



ESRS & BRSR

Regulatory formats: the EU's ESRS under CSRD, and SEBI's BRSR for India's top 1,000 listed companies.

Three Materiality Lenses

The single biggest difference between frameworks is which question they ask about each topic.

Impact Materiality (GRI)	How does the organisation affect the economy, environment, and people, whether or not it affects profit? Audience: all stakeholders.
Financial Materiality (ISSB)	Which sustainability-related risks and opportunities could affect the company's prospects and enterprise value? Audience: investors and lenders.
Double Materiality (ESRS)	Both lenses together: a topic is material if it passes either test. The EU's 2026 Omnibus kept this principle while cutting datapoints.

IFRS S1 and S2: The Investor Baseline

Issued in June 2023 by the International Sustainability Standards Board, effective from 1 January 2024 wherever a jurisdiction adopts them.

Four Pillars (from TCFD)	IFRS S1 asks for governance, strategy, risk management, and metrics and targets for sustainability-related risks and opportunities.
IFRS S2 for Climate	Adds Scope 1, 2 and 3 GHG emissions measured under the GHG Protocol, climate resilience and scenario analysis, and industry-based metrics.
SASB Standards	Industry-specific metrics for 77 industries, now maintained by the ISSB, make disclosures comparable within a sector.
Reported With the Financials	Sustainability information is reported at the same time and for the same period as the financial statements, tying it to financial-reporting discipline.

GRI Standards: The Impact Lens

The most widely used voluntary standards, organised as universal, topic, and sector standards.

Universal Standards (2021)	GRI 1 (foundation), GRI 2 (general disclosures), and GRI 3 (material topics). In force since 1 January 2023, with impact materiality as the organising idea.
Topic Standards	GRI 101 Biodiversity (effective 1 January 2026), plus GRI 102 Climate Change and GRI 103 Energy, published June 2025 and effective 1 January 2027, replacing GRI 305 and GRI 302.
What GRI 102 Adds	Emissions across Scopes 1–3, targets, transition plans, and impacts on workers and communities (just transition).
Working Together	In June 2025 GRI and the IFRS Foundation said organisations can use GRI 102 with IFRS S2 to report on both impacts and financial risks and opportunities.

How Far Along Is the World?

28

Jurisdictions that had adopted ISSB Standards on a voluntary or mandatory basis by April 2026, with 12 more planning to (S&P Global tracking)

36–39

Jurisdictions the IFRS Foundation counted in 2025 as using, or taking steps toward, ISSB Standards

≈ 40%

Estimated alignment of India's BRSR with IFRS S2, per an ORF analysis; BRSR was built around TCFD and GRI

≈ 60%

Cut in mandatory ESRS datapoints under the EU's revised standards adopted in July 2026

Convergence is real but uneven: jurisdictions adopt, adapt, or simplify. Expect "collect once, report many" to remain the practical rule for years.

Sources: S&P Global ISSB tracker (April 2026); IFRS Foundation jurisdictional counts (2025); ORF (BRSR interoperability); EU revised ESRS (July 2026).

◆ INTERACTIVE

CASE STUDY

Simplification vs. Comparability: The EU's 2025–26 Reset

Omnibus I Directive (EU) 2026/470 and the revised ESRS

After the first wave of CSRD reporting, the EU cut the scope of mandatory reporting and rewrote its standards. The Directive was adopted on 24 February 2026 and entered into force on 18 March 2026.

Who Reports Now

Only companies with more than 1,000 employees and more than €450 million net turnover (previously about 250 employees or €50 million). The Commission estimated roughly 80% fewer companies in scope.

The Standards

The revised ESRS, adopted on 3 July 2026, cut mandatory datapoints by roughly 60%, drop sector-specific standards, and apply from financial year 2027.

What Survived

Double materiality remains the foundation: in-scope companies must still assess both impact and financial materiality.

The Trade-off

Supporters point to lower burden. Critics worry that fewer reporters and fewer datapoints reduce comparability and coverage. Both effects are real; weigh them.

BRSR, BRSR Core and Assurance in India

SEBI’s framework is mandatory, phased, and increasingly assured.

BRSR	Mandatory since FY 2022–23 for the top 1,000 listed companies by market capitalisation, filed with the annual report and structured around the nine principles of the National Guidelines on Responsible Business Conduct.
BRSR Core	A focused set of KPIs under nine attributes. Assessment or assurance phases in from the top 150 (FY 2023–24) to the top 1,000 (FY 2026–27); in December 2024 SEBI allowed a choice between “assessment” and “assurance”.
Assurance Standard	ICAI’s Standard on Sustainability Assurance (SSA 5000), modelled on the global ISSA 5000, was issued in 2026 with a reported effective date of 1 April 2027.
Open Gaps	An ORF analysis estimates BRSR is about 40% aligned with IFRS S2 and notes that covering only the top 1,000 firms captures less than half of India’s emissions footprint.

◆ INTERACTIVE

MATCH THE QUESTION TO THE FRAMEWORK

Which Framework Answers This?

For each question, choose the framework that is built to answer it.

1	How does our company affect communities and ecosystems, regardless of financial effect?
2	Which climate risks could change our enterprise value, and how are they governed?
3	How do we count Scope 1, 2 and 3 emissions consistently?
4	Is our 2030 emissions target consistent with a 1.5°C pathway?
5	Which industry-specific metrics should a steel company disclose to investors?
6	How do we report the impacts of our energy use (GRI 103)?

GRI	ISSB	GHG PROT.	SBTi
1	2	3	4
6	5		

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

Which materiality concept do the ISSB standards use?

A Impact materiality only

B Financial materiality: what could affect enterprise value and investor decisions ✓

C Double materiality

D None; ISSB has no materiality concept

ANSWER & WHY

B. Financial materiality: what could affect enterprise value and investor decisions

The ISSB is investor-focused: it asks which sustainability-related risks and opportunities could affect the company's prospects. GRI uses impact materiality; ESRS uses double materiality.

SECTION 3 OF 4

From Data to Disclosure

Reporting is a data process before it is a document. The controls decide whether it can be trusted.

Six Steps From Boundary to Published Disclosure

The cycle repeats every year, and each step leaves evidence the next one relies on.

1. Scope & Boundary	Decide the reporting entity, period, and which operations and value-chain parts are in.
2. Materiality	Identify which topics matter (impact, financial, or both) and document the method.
3. Collect & Calculate	Gather activity data and apply emission factors and conversion rules consistently.
4. Control & Review	Reconcile to invoices and meters, test for anomalies, and keep an audit trail.
5. Assure	Independent limited or reasonable assurance on selected disclosures.
6. Disclose & Tag	Publish in the required framework and format, with machine-readable tagging where required.

SECTION 3 · WORKED EXAMPLE

A Materiality Screen Under Three Lenses

GIVEN

- Four topics scored 1–5 for (impact, financial): GHG emissions (5, 4); water use (4, 2)
- Employee safety (3, 3); data privacy (2, 4)
- Threshold: a topic is material on a lens if its score is 3.5 or higher

FORMULA

Impact lens: Impact ≥ 3.5 · Financial lens: Financial ≥ 3.5
Double materiality: either test passes

STEP-BY-STEP CALCULATION

1. Impact lens (GRI-style): GHG emissions (5) and water use (4) are material
2. Financial lens (ISSB-style): GHG emissions (4) and data privacy (4) are material
3. Double materiality = union: GHG emissions, water use, and data privacy (3 of 4 topics)
4. Employee safety (3, 3) passes neither lens at this threshold

3 of 4 Topics

The lens changes the list: water is material only to impact; data privacy only to finance. Documenting the thresholds and scoring method is what makes the screen auditable. (Scores are illustrative.)

Setting and Holding the Boundary

Numbers are only comparable if the boundary they describe stays stable and disclosed.

Organisational Boundary	Equity share, financial control, or operational control (from our footprint lecture). Pick one, apply it consistently, and disclose it.
Operational Boundary	Which scopes and Scope 3 categories are in. Excluded categories should be named, with reasons.
Alignment With Financial Reporting	ISSB asks for sustainability information for the same reporting entity and period as the financial statements, so boundaries must reconcile.
Consistency Over Time	Changes in boundary or method break comparability, which is why base-year recalculation rules exist.

SECTION 3 · WORKED EXAMPLE

Base-Year Recalculation After an Acquisition

GIVEN

- Base year 2020 emissions: 100,000 tCO_2e; company policy: recalculate if a structural change exceeds 5%
- In 2024 the company acquires a plant that emitted 12,000 tCO_2e in 2020
- 2024 emissions, including the new plant: 104,000 tCO_2e

FORMULA

Significance = Structural change ÷ Base-year emissions
Recalculated Base = Original Base + Acquired Emissions

STEP-BY-STEP CALCULATION

1. Significance: 12,000 ÷ 100,000 = 12%, above the 5% threshold, so recalculate
2. Recalculated base year: 100,000 + 12,000 = 112,000 tCO_2e
3. Against the unadjusted base: 104,000 vs 100,000 looks like +4%
4. Against the recalculated base: 104,000 vs 112,000 is −7.1%, the like-for-like change

−7.1%, Not +4%

Without recalculation, the acquisition makes a real reduction look like an increase. The rule keeps the comparison like for like. (Figures are illustrative.)

What Makes Sustainability Data Auditable

Assurance practitioners test the process as much as the number.

Primary Before Secondary	Use metered or invoiced data where possible; estimate with emission factors only where measurement is impractical, and label estimates as such.
Versioned Methods	Record the emission-factor source and year (for example the CEA grid factor) so any number can be reproduced later.
Reconciliation	Tie totals to invoices, meters, or the general ledger, and investigate unexplained variances before publishing.
Restatement Discipline	When an error or method change is found, restate prior years and disclose why, rather than quietly changing the series.

SECTION 3 · YOUR DOMAIN

The ESG Data Pipeline: From ERP to Tagged Disclosure

Reporting is a data-engineering problem, which makes it a computer science opportunity.



Ingest

Utility bills, smart meters and IoT feeds, ERP and procurement data, and supplier portals, each with its own format and cadence.



Calculate

A versioned emission-factor library, boundary rules, and unit tests so a calculation can be rerun and reproduced.



Control

Reconciliations, anomaly detection, maker-checker review, and a lineage record from source document to reported number.



Publish

Mapping one dataset to GRI, ISSB, ESRS and BRSR, with machine-readable tagging (XBRL) where regulators require it.



AI Assistance

Language models can extract and classify data, but every figure needs provenance and human review before assurance.

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

In the base-year example, why was the 2020 base year recalculated?

A Because emissions rose in 2024

B Because the acquisition changed the company’s structure by 12%, above the 5% significance threshold ✓

C Because the grid emission factor changed

D Because regulators require recalculation every year

ANSWER & WHY

B. Because the acquisition changed the company’s structure by 12%, above the 5% significance threshold

A structural change above the company’s significance threshold requires restating the base year so later years are compared like for like. A 12% change against a 5% threshold triggers it.

SECTION 4 OF 4

Credibility & Assurance

A disclosure is only as valuable as a reader's reason to believe it.

Limited vs. Reasonable Assurance

Limited Assurance

The practitioner mainly makes inquiries and applies analytical procedures; the conclusion is stated negatively ("nothing has come to our attention…").

A lower level of assurance: quicker and less costly.

A common starting point when a new reporting regime begins.

Risk: readers over-read "assured" as "audited".

Reasonable Assurance

The practitioner gathers extensive evidence, tests controls, and samples data; the conclusion is stated positively ("in our opinion, fairly stated").

A higher level of assurance, closer to a financial audit: slower and costlier.

The higher bar. India's BRSR Core was framed around it, though SEBI later allowed an "assessment" option.

Risk: the scope may cover only selected disclosures, not the whole report.

ISSA 5000 and India's SSA 5000

Assurance is getting its own global standard.

ISSA 5000	Published by the IAASB in November 2024: the first comprehensive global standard for sustainability assurance, covering limited and reasonable assurance on any sustainability topic or framework.
Effective Date	For periods beginning on or after 15 December 2026, wherever a jurisdiction adopts it. It replaces ISAE 3410 for greenhouse-gas assurance when it takes effect.
India	ICAI's SSA 5000, based on the international standard, is reported to take effect from 1 April 2027 for the top 1,000 listed companies, and for voluntary use by others.
What It Means for Reporters	Strengthen internal controls over ESG data now: weak controls turn into modified assurance conclusions.

Four Dates That Reshape Reporting

FY 2026–27

BRSR Core assessment or assurance reaches the top 1,000 Indian listed companies.

15 Dec 2026

ISSA 5000 applies to sustainability information for periods beginning on or after this date.

1 Jan 2027

GRI 102 Climate Change and GRI 103 Energy take effect; revised ESRS apply to in-scope EU companies from FY 2027.

2027–28

Revised GHG Protocol Scope 2 expected in 2027, implementation likely from 2028; SBTi Net-Zero Standard V2 mandatory for new targets from 1 Jan 2028.

SECTION 4 · RED FLAGS

Four Warning Signs in a Sustainability Report

Use these as a reader, and as a preparer checking your own draft.

Moving Goalposts	The base year, boundary, or method changes without disclosure or restatement, so the trend cannot be checked.
Intensity-Only Storytelling	Intensity improves while absolute totals are omitted or rising (our worked example).
Selective Coverage	Scope 3 or the largest impacts are missing, while minor ones are reported in detail.
Assurance Ambiguity	A report says “verified” without saying who, to what level, and which disclosures were covered.

SECTION 4 · WORKED EXAMPLE

Why the Scope 2 Revision Matters: Annual vs. Hourly Matching

GIVEN

- A data centre uses 10,000 MWh a year and buys solar certificates equal to 10,000 MWh
- Solar output covers only 60% of consumption in the hours it is actually used (illustrative)
- Grid emission factor for unmatched hours: 0.7 tCO_2e per MWh (illustrative)

FORMULA

Market-based Emissions = Unmatched Consumption × Grid Factor

STEP-BY-STEP CALCULATION

1. Annual matching (today): certificates cover 100% of annual use, so market-based Scope 2 = 0 tCO_2e
2. Hourly matching (proposed): unmatched share = 1 − 0.60 = 40%
3. Unmatched consumption: 10,000 × 0.40 = 4,000 MWh
4. Emissions: 4,000 × 0.7 = 2,800 tCO_2e

$0 \rightarrow 2{,}800\ tCO_2e$

The GHG Protocol's 2025 consultation proposed hourly matching and deliverability for market-based claims. Still under revision: a final standard is expected in 2027, with implementation likely from 2028.

◆ INTERACTIVE

THINK & SHARE

“Your board wants to announce that emissions intensity fell 20% this year. Total emissions rose 4%. What must the report say alongside the headline?”

Think about absolute figures, boundary, base year, and the level of assurance.

⏱ 3 MIN

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

From when does the IAASB's ISSA 5000 apply, in jurisdictions that adopt it?

A Periods beginning on or after 1 January 2024

B Periods beginning on or after 15 December 2026 ✓

C Periods beginning on or after 1 January 2030

D It has already been withdrawn

ANSWER & WHY

B. Periods beginning on or after 15 December 2026

ISSA 5000 was published in November 2024 and is effective for assurance on sustainability information for periods beginning on or after 15 December 2026 (early application permitted).

BRINGING IT TOGETHER

Collect Once, Report Many: An Approximate Crosswalk

One metric family can feed several frameworks. Check each framework's current version before relying on a mapping.

GHG Emissions	GRI 102 · IFRS S2 · ESRS E1 · BRSR Core greenhouse-gas footprint
Energy	GRI 103 · ESRS E1 · BRSR Core energy footprint
Water	GRI 303 · ESRS E3 · BRSR Core water footprint
Waste & Circularity	GRI 306 · ESRS E5 · BRSR Core circularity
Workforce Safety	GRI 403 · ESRS S1 · BRSR Core wellbeing and safety

BRINGING IT TOGETHER

A Reporting Readiness Checklist

Before the first number is published, you should be able to answer yes to each of these.

Metrics Defined	Every metric has a unit, formula, owner, and source, and passes the five tests.
Materiality Documented	The method, thresholds, and results are written down and approved.
Boundary & Base Year Set	Entity, scopes, base year and recalculation policy are disclosed and consistent.
Controls in Place	Reconciliations, lineage, versioned factors, and review are operating and evidenced.
Assurance-Ready	The level and scope of assurance are agreed, and the evidence file is organised.
Framework Mapping Done	Each disclosure maps to the frameworks the company must or chooses to follow.

◆ INTERACTIVE

APPLY IT ALL

“Open the latest BRSR or sustainability report of any large Indian listed company. Within three minutes, find its boundary, its base year, and whether its data is assured. What is missing?”

This is how an analyst or auditor starts a review.

⏱ 4 MIN